

The Crouzeix Conjecture Mentioned in arXiv:1711.06042

Directly answered by arXiv:2608.03841

Literature-status identification

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Source statement

In the Introduction of P. Gorkin and J. R. Partington, *Norms of truncated Toeplitz operators and numerical radii of restricted shifts*, arXiv:1711.06042 (printed page 2), the extracted open-problem signal is the classical Crouzeix conjecture:

$$\|p(A)\| \leq 2 \max\{|p(z)| : z \in W(A)\} \quad (\text{C})$$

for every polynomial p and every matrix A . The paper says that this question remained open at the time of writing. It does not pose a narrower new conjecture about truncated Toeplitz operators at this location.

Direct later answer

E. Lorist and F. L. Schwenninger, *A solution to Crouzeix's conjecture*, arXiv:2608.03841v1 (August 4, 2026), states in Theorem 3:

Let A be a bounded operator on a Hilbert space H . Then $W(A)$ is a 2-spectral set; that is,

$$\|f(A)\| \leq 2 \sup_{z \in W(A)} |f(z)| \quad (\text{LS})$$

for all rational functions f with poles off $\overline{W(A)}$.

Taking $H = \mathbb{C}^d$ and $f = p$ gives (C) exactly. Thus the question cited in arXiv:1711.06042 is directly answered affirmatively, and the later theorem is strictly stronger in both its operator and function scope.

Proof identification and status

The resolving paper supplies a proof, not merely a reformulation. Its new Lemma 1 is a perturbation result for 2-dilations: if

$$E_n = 2V^*Q^{*n}V - T^{*n}$$

is uniformly bounded and commutes with T , where Q is a contraction and V an isometry, then $\|T\| \leq 2$. For $T = f(A)$, the double-layer-potential representation provides Q, V , and the commuting errors E_n ; Lemma 1 then yields (LS).

This packet records a direct literature resolution. It does not claim a new proof. The answer is currently a very recent arXiv preprint, not a peer-reviewed publication; confidence is based on the exact theorem/proof match and on an independent contemporary proof cited by the authors. The unrestricted completely bounded variant is explicitly outside the theorem's scope, but that limitation does not affect the scalar matrix conjecture stated in arXiv:1711.06042.

References

- [1] P. Gorkin and J. R. Partington, *Norms of truncated Toeplitz operators and numerical radii of restricted shifts*, arXiv:1711.06042.
- [2] E. Lorist and F. L. Schwenninger, *A solution to Crouzeix's conjecture*, arXiv:2608.03841v1, 2026.